

Sierra Thuy-Dan Vo

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PROFESSIONAL SUMMARY

Data scientist and analytics professional with a background in statistics, finance, and applied machine learning. Experienced in SQL, Python, forecasting, model evaluation, and business-facing analytics across fintech, product metrics, revenue operations, and customer behavior.

TECHNICAL SKILLS

Programming & Data: Python, SQL, R, pandas, NumPy, MySQL, Excel/VBA, Git

Machine Learning & Statistics: scikit-learn, PyTorch, TensorFlow, LightGBM, XGBoost, predictive modeling, regression, classification, time-series forecasting, feature engineering, model evaluation

Analytics & Visualization: Tableau, Power BI, NLP, computer vision, decision support, LaTeX

PROFESSIONAL EXPERIENCE

Co-Founder Jul 2024 – Present

Babywearco LLC New York, NY

- Operated an e-commerce business, managing product selection, pricing, inventory, marketing, and customer acquisition
- Analyzed conversion metrics, customer behavior, and advertising performance to evaluate growth and marketing efficiency
- Built reporting processes to track sales, inventory turnover, and key business metrics

Senior Financial Data Analyst / Accounts Receivable Team Lead Jul 2022 – Dec 2024

By Design LLC New York, NY

- Automated invoicing, reducing aged accounts receivable by 12% and doubling collection rates
- Used multiple regression to identify unprofitable customers, increasing profitability by 7%
- Automated Python reports and linear-regression forecasts, cutting administrative work by 50%

Quantitative Analyst Mar 2022 – Jul 2022

UBS Group AG – Quantitative Analysis and Data Science Team New York, NY

- Queried and joined SQL data and automated Python time-series revenue forecasts, reducing quarterly work by 1 week
- Built a Python pipeline combining SQL data and scraped text to generate dynamic wealth manager profiles
- Built regression and tree-model decision metrics predicting a 60% wealth-manager bring-over rate
- Delivered strategic insights to management through Tableau and Power BI dashboards

Machine Learning Researcher Nov 2020 – Jul 2021

Columbia University – Graduate School of Arts and Sciences New York, NY

- Reimplemented a multimodal emotion-recognition workflow combining article text, engineered facial-landmark features, and pixel-based CNN predictions for media analytics
- Built article/image scraping and model-aggregation pipelines; accompanying research showed that combining landmark and pixel-model outputs improved 8-class emotion classification accuracy from about 58% to 76.7%

Data Science Researcher Jun 2019 – Dec 2019

Case Western Reserve University – School of Medicine Cleveland, OH

- Analyzed structured and unstructured medical data for 10,000 patients over 16 years using R
- Identified brain-tumor risk factors and treatment outcomes using logistic regression and survival analysis
- Presented findings in weekly reports and contributed to a peer-reviewed manuscript

SELECTED PROJECTS

Crypto Hybrid Forecasting Model | PyTorch, Time Series, Graph Neural Nets, Model Evaluation

- Built an end-to-end PyTorch pipeline for 14-asset cryptocurrency forecasting, comparing LSTM, GCN, and hybrid graph-sequence models with technical indicators, leakage-aware preprocessing, baseline benchmarks, and adjacency ablations
- Used RMSE, MAE, directional accuracy, and per-asset error analysis to show that added graph-model complexity was not justified under the tested setup

Music Revenue Forecasting & Marketing Allocation | Python, LightGBM, Forecasting, Marketing Analytics

- Built an end-to-end revenue forecasting pipeline that aggregated artist-month sales data, engineered lagged and rolling time-series features, and compared baseline, linear, LightGBM, CatBoost, and ensemble models
- Improved RMSE by approximately 35% over the naive baseline and translated forecasts into marketing-budget recommendations using predicted revenue, growth, efficiency, and volatility scoring

EDUCATION

Columbia University New York, NY

M.A., Statistics, Concentration in Machine Learning Aug 2020 – Dec 2021

Case Western Reserve University Cleveland, OH

M.S., Finance – Big Data Analytics | B.S., Finance & Statistics – Actuarial Science Aug 2016 – Aug 2020